

Math 421

Wednesday, September 24

Announcements

- Homework 3 due Friday
- Grace's Drop-In Hours
 - **Today:** 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4

Activity & Discussion Question

Find the supremum/infimum of the following sets (if they exist).

1. (a, b) $\inf: a$
 $\sup: b$

2. $\{n \in \mathbb{N} : n^2 < 3\}$ $\inf: 1 = \min$
 $\mathbb{N} = \{1, 2, \dots\}$ $\sup: 1 = \max$

3. $\{x \in \mathbb{R} : x^2 < 3\}$ $\sup: \sqrt{3}$
 $\inf: -\sqrt{3}$
 $(-\sqrt{3}, \sqrt{3})$

4. $\{\sin \frac{n\pi}{4} : n \in \mathbb{N}\}$ $\sup: 1 = \max$
 $\inf: -1 = \min$
 $\subset \mathbb{R}$

❖ Is the empty set bounded?

• $\forall x \in \emptyset, x \leq M \quad M \in \mathbb{R}$
 $\forall x \in \emptyset, x \geq m \quad m \in \mathbb{R}$

• No supremum & infimum.

$\{\sin \frac{n\pi}{4} : n \in \mathbb{N}\} =$

$\{\sin \frac{\pi}{4}, \sin \frac{2\pi}{4}, \sin \frac{3\pi}{4}, \dots\} =$

$\{\frac{\sqrt{2}}{2}, 1, 0, -\frac{\sqrt{2}}{2}, -1\}$

Completeness Axiom

Completeness Axiom. Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound. In other words, $\sup S$ exists and is a real number.

Q: Does the completeness axiom hold for $\mathbb{Q}$? No.

$$\{r \in \mathbb{Q} : 0 \leq r \leq \sqrt{2}\} \subseteq \mathbb{R} \rightarrow \sup: \sqrt{2}$$

$$\{r \in \mathbb{Q} : r^2 \leq 2\} \subseteq \mathbb{Q} \rightarrow \sup \text{ DNE.}$$

Corollary: Every nonempty subset S of $\mathbb{R}$ that is bounded below has a greatest lower bound.

Unbounded Sets & Notation

Q: What does it mean for a set to be **unbounded**?

- Not bounded above: There is no upper limit.

OR There is no value $M \in \mathbb{R}$ s.t. $M \geq s$ for all $s \in S$.

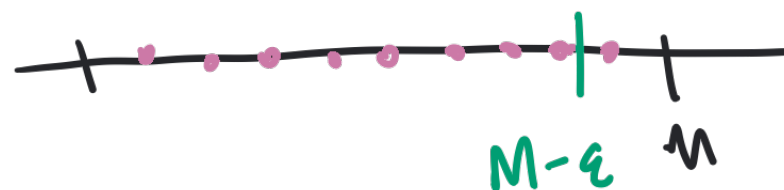
OR for all $M \in \mathbb{R}$ there exists $s_0 \in S$ s.t. $s_0 > M$.

- $\sup(S) = \begin{cases} \text{least upper bound} & S \neq \emptyset \text{ and bounded above} \\ \infty & S \neq \emptyset \text{ and not bounded above.} \end{cases}$

$$M - \varepsilon < s \leq M$$

Equivalent Definition of $\sup S$

Thm. Let $M \in \mathbb{R}$ be an upper bound for a set $S \subseteq \mathbb{R}$. Then $M = \sup S$ if and only if for every $\varepsilon > 0$ there ^{exists} ~~exists~~ some $s \in S$ with $M - \varepsilon < s$.



Discussion Questions:

- ❖ Describe what this theorem is saying in your own words.
- ❖ Convince yourselves that this theorem is true, it might be helpful to think in terms of a specific example and then see how it can generalize.

Archimedean Property

Thm. Let $a, b \in \mathbb{R}$. If $a > 0$ and $b > 0$, then for some positive integer n , we have $na > b$. OR $a > \frac{b}{n}$.

Discussion Questions:

- ❖ What is this theorem saying in your own words?
- ❖ What does this property say if $a = 1$? $n > b$
- ❖ What does this property say if $b = 1$? $a > \frac{1}{n}$

Archimedean Property

Thm. Let $a, b \in \mathbb{R}$. If $a > 0$ and $b > 0$, then for some positive integer n , we have $a > \frac{b}{n}$.

• Use proof by contradiction: $a, b \in \mathbb{R}, a > 0, b > 0$
for all $n \in \mathbb{N}$ $a \leq \frac{b}{n} \Rightarrow n \leq \frac{b}{a}$

Discussion.

• Using Completeness Axiom: $\mathbb{N}$ is nonempty, bounded above by $\frac{b}{a}$. So $\sup \mathbb{N}$ exists.

$$n \in \mathbb{N} \rightarrow n+1 \in \mathbb{N} \Rightarrow n+1 \leq \sup \mathbb{N} \Rightarrow n \leq \underbrace{\sup \mathbb{N} - 1}_{\text{upper bound}}$$

contradiction.

Archimedean Property

Proof. Suppose by way of contradiction that for $a, b \in \mathbb{R}$, $a > 0$, $b > 0$ we have $a \leq \frac{b}{n}$ for all $n \in \mathbb{N}$. Then $n \leq \frac{b}{a}$ for all $n \in \mathbb{N}$, so $\frac{b}{a}$ is an upper bound for $\mathbb{N}$. $\mathbb{N}$ is non-empty and bounded above, so by the completeness Axiom, $\sup \mathbb{N} \in \mathbb{R}$ exists. Note that $n \in \mathbb{N} \Rightarrow n+1 \in \mathbb{N}$ so $n+1 \leq \sup \mathbb{N} \Rightarrow n \leq \sup \mathbb{N} - 1$. So $\sup \mathbb{N} - 1$ is an upper bound for $\mathbb{N}$, but $\sup \mathbb{N} - 1 < \sup \mathbb{N}$ which contradicts the fact that $\sup \mathbb{N}$ is the least upper bound.